



Aditya Vikram Manocha

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Education

Manipal Institute of Technology, Manipal 2024 – 2028
B.Tech in Computer Science Engineering — CGPA: 7.18 Reg No: 240905562

Experience

Ashland May 2026 – Present
Data Scientist Intern

- Engineered AI-assisted workflow automations using **Microsoft Copilot Studio**, cutting manual effort in repetitive enterprise operations and improving process throughput.
- Led structured market intelligence and competitive analysis efforts, turning multi-source data acquisition into actionable insights for business strategy and decision-making.
- Built automation pipelines integrating AI tools for reporting and information retrieval, reducing turnaround time on recurring operational workflows.

Projects

MAFE – Multi-Agent Feature Engineering Framework Completed
Python, Scikit-learn, Streamlit, Multi-Agent Systems — github.com/HEX-CLOUD/MAFE

- Architected a research-oriented multi-agent framework with specialized agents for feature transformation, interaction discovery, redundancy pruning, and leakage detection in tabular ML pipelines.
- Designed a coordinator-driven orchestration layer to automate feature engineering while constraining feature-space explosion and preserving model interpretability.
- Benchmarked the framework against baseline approaches via ablation studies on the Adult Income dataset, demonstrating consistent gains in downstream predictive performance.

AI Hiring Copilot In Progress
FastAPI, LangChain, LangGraph, OpenAI/Gemini API, PostgreSQL, ChromaDB, Docker

- Building an AI recruiting copilot with FastAPI and LangGraph to automate resume parsing, job-description matching, and explainable candidate ranking.
- Designing a vector-search layer over ChromaDB to power semantic resume retrieval and automated interview-question generation.
- Architecting a recruiter-facing dashboard backed by PostgreSQL for structured candidate data and ranking audit trails, containerized with Docker for reproducible deployment.

Multi-Agent Research Assistant In Progress
LangGraph, CrewAI, FastAPI, Tavily, Vector Database

- Developing a multi-agent research pipeline – research, fact-checking, citation, and report-generation agents – orchestrated with LangGraph and CrewAI.
- Integrating the Tavily search API for autonomous, tool-using web research grounded in retrieved sources.
- Building a vector-database-backed retrieval layer to support structured, citation-linked report generation.

RecruitOCR Completed
Python, OpenCV, Tesseract OCR, Streamlit, Google Sheets API — github.com/HEX-CLOUD/Recruitool

- Engineered an OCR pipeline converting unstructured resume PDFs into ATS-ready structured candidate records.
- Built an interactive Streamlit dashboard for automated resume ingestion, validation, and recruiter-facing candidate tracking, synced via the Google Sheets API.

Customer Churn Prediction Completed
Python, TensorFlow, Scikit-learn, Pandas — github.com/HEX-CLOUD

- Developed an ANN-based churn prediction pipeline to flag high-risk customers from behavioral and transactional features, evaluated via Accuracy, Precision-Recall, and ROC-AUC.

Technical Skills

- Languages:** Python, C++, C, Java, JavaScript, SQL
- Frameworks & Tools:** FastAPI, Streamlit, Docker, Git, GitHub, Linux, Microsoft Copilot Studio, Jupyter Notebook
- AI/ML:** TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas
- LLM & Agentic AI:** LangChain, LangGraph, CrewAI, OpenAI/Gemini APIs, RAG, Vector Search (ChromaDB)
- Databases:** PostgreSQL, MongoDB, ChromaDB, SQL
- Core CS:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Achievements & Certifications

- Qualified Smart India Hackathon (SIH) Round 1 – Top 40 institute teams
- Solved 150+ problems on LeetCode
- Supervised Machine Learning – Andrew Ng (Coursera)